Discover then Refine: A Joint Multiple Choice Learning and Flow Matching Framework for Heat Demand Forecasting

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Abstract

District heating optimization requires forecasting models that characterize the inherent multimodality of demand. While generative approaches address this, they typically rely on heavy, undirected sampling and require post-hoc clustering to yield actionable results. We propose a two-stage framework integrating Multiple Choice Learning (MCL) with Flow Matching to generate forecasts with explicit probabilities. Our approach offers three primary benefits: First, MCL identifies diverse hypotheses that guide targeted sampling, ensuring comprehensive coverage of the outcome space, including low-probability events. Second, because these hypotheses act as an aligned prior, the flow model requires fewer integration steps, reducing inference time compared to global generative baselines. Finally, by providing explicit probabilities for each hypothesis, the framework equips operators with a set of weighted scenarios for risk assessment. Evaluated on real-world public data from 3,021 buildings in Aalborg, Denmark, our method achieves competitive accuracy and calibration while allowing for real-time operational decision-making.

1. Introduction

District Heating Networks (DHNs) are critical for sustainable urban thermal regulation (Angelidis et al., 2023), yet their efficiency depends on accurate heat demand forecasting to balance production and load. Forecasting is inherently difficult due to non-stationary consumption driven by weather and stochastic human behavior (Huang et al., 2023). Research for heat demand forecasting primarily focused on deterministic predictions, from tree-based ensembles (Gong et al., 2020) to Spatio-Temporal GNNs (Wang et al., 2023; Huang et al., 2025), these deterministic models often fail by outputting an unlikely average state rather than distinct physical regimes. To address multimodality, generative models like diffusion and flow-based methods have been adopted for timeseries (Rasul et al., 2021; Tashiro et al., 2021; Kollovieh et al., 2025; Miraki et al., 2025). However, as highlighted by the *Probabilistic Scenarios* paradigm (Dai et al., 2026), sampling-based approaches can face critical limitations in real-time energy management: (i) *Probability Absence*, lacking explicit weights for modes; (ii) *Coverage Inadequacy* for high-risk tail events; and (iii) *Inference Cost* due to heavy sampling and multiple denoising steps.

We propose **MC-Flow**, a hierarchical framework that bridges the gap between discrete hypothesis learning and continuous generative modeling. We leverage Multiple Choice Learning (Guzmán-rivera et al., 2012; Letzelter et al., 2024) to predict a discrete set of K distinct future trajectories. Extending on prior work levereging MCL for timeseries (Cortés et al., 2025)), we treat the hypotheses as a structured priors for a Conditional Flow Matching (CFM) (Tong et al., 2024) module. This allows us to decouple global structural uncertainty from local variance, ensuring each generated scenario inherits an explicit probability while allowing efficient low step count during inference.

2. Methodology

2.1. System Overview

The objective of our proposed framework, MC-Flow, is to map a sequence of historical observations and exogenous covariates into a set of continuous, plausible future demand trajectories. To achieve this, MC-Flow decomposes the forecasting task into two sequential stages: (1) a discrete hypothesis discovery stage that identifies K distinct target modalities, and (2) a continuous refinement stage that models the local variance within each mode via conditional flow matching. For a visual diagram of MC-Fow, we refer to Appendix D.

2.2. Problem Formulation and Context Encoding

Let $\mathbf{y}_p \in \mathbb{R}^P$ denote the sequence of univariate historical heat demand observations over P time steps. Let

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$\mathbf{C} \in \mathbb{R}^{(P+F)\times C}$ represent the exogenous covariates available across both the historical window and the future prediction horizon F, where C is the feature dimension. These covariates comprise normalized external temperature, continuous calendar encodings (sine and cosine transformations of day-of-week and day-of-year), and a binary observation indicator. This indicator equals 1 during the historical window of length P and 0 during the forecasting horizon of length F, explicitly conditioning the encoder on the temporal boundary between the known past and the unknown future. Additionally, let $\mathbf{m}$ denote the static metadata associated with each specific building (comprising its learned ID embedding, unit type, and normalized construction year). The objective is to forecast the future target trajectory $\mathbf{y}_f \in \mathbb{R}^F$.

To capture temporal dynamics and external drivers, the historical demand $\mathbf{y}_p$ and covariates $\mathbf{C}$ are processed by a Context Encoder to extract a time-pooled global context vector $\mathbf{h}_{ctx}$. This vector serves as the foundational condition for the entire pipeline: it provides the initial hidden state for the sequence-to-sequence generation, acts as the input feature for discrete regime classification, and serves as an additional static condition for the flow matching network.

2.3. Discrete Regime Discovery via Multiple Choice Learning

This stage maps the input context space to a discrete set of K candidate scenarios alongside their associated probabilities, establishing a mapping $\Phi : \mathcal{C} \to \{(\mu_k, \sigma_k^2, \pi_k)\}_{k=1}^K$. This is driven by two parallel operations:

Hypothesis Generation: Conditioned on the global context $\mathbf{h}_{ctx}$, covariates $\mathbf{C}_f$, and a learned mode embedding, the Hypothesis Generator employs a shared Sequence-to-Sequence GRU architecture. For each hypothesis $k \in \{1, \ldots, K\}$, it outputs a structural mean $\mu_k \in \mathbb{R}^F$, representing the expected shape of the demand regime, and a variance parameter $\sigma_k^2 \in \mathbb{R}^F$, representing the expected prediction error.

Regime Classification: In parallel, a Probability Head projects $\mathbf{h}_{ctx}$ into K logits and applies a Softmax to yield occurrence probabilities π_k. To train this classifier, we derive soft targets y_{soft} by applying a Softmax over the negative Mean Squared Error (MSE) of all K hypotheses, scaled by a temperature parameter τ_{target}. This formulation provides a richer gradient signal, smoothly aligning the predicted prior distribution with the empirical performance of the Hypothesis Generator.

To maintain mode separation and prevent hypothesis collapse, the generator is trained using a Relaxed Winner-Takes-All (R-WTA) objective. For a given ground truth $\mathbf{y}_f$, we first identify the winning hypothesis k^*:

$$k^* = \arg \min_{k\in\{1,\ldots,K\}} \|\mu_k - \mathbf{y}_f\|_2^2 \tag{1}$$

R-WTA assigns a weight q_k to each head, where the winner receives the majority of the gradient and non-winners receive a small update ϵ for exploration:

$$q_k = \begin{cases} 1-\epsilon & \text{if } k = k^* \\ \frac{\epsilon}{K-1} & \text{if } k \neq k^* \end{cases} \tag{2}$$

The complete optimization objective $\mathcal{L}_{\text{discrete}}$ is composed of three parts:

1. **Structural Mean Loss:** Penalizes the distance between the hypotheses and the ground truth:

$$\mathcal{L}_\mu = \sum_{k=1}^K q_k \|\mu_k - \mathbf{y}_f\|_2^2 \tag{3}$$

2. **Variance Regression Loss:** Drawing on Direct Epistemic Uncertainty Prediction (DEUP) (Lahlou et al., 2022), this trains σ_k^2 to estimate the empirical squared error:

$$\mathcal{L}_\sigma = \sum_{k=1}^K q_k \left\| \sigma_k^2 - \|\mu_k - \mathbf{y}_f\|_2^2 \right\|_2^2 \tag{4}$$

3. **Regime Classification Loss:** A cross-entropy term for the predicted logits:

$$\mathcal{L}_\pi = \text{CE}(\pi/\tau, y_{\text{soft}}) \tag{5}$$

The total objective is the summation of these components:

$$\mathcal{L}_{\text{discrete}} = \mathcal{L}_\mu + \mathcal{L}_\sigma + \mathcal{L}_\pi \tag{6}$$

2.4. Mode-Conditioned Gaussian Priors

The structural parameters $\{(\mu_k, \sigma_k^2)\}_{k=1}^K$ define a discrete set of conditional Gaussian priors over the future prediction horizon. During training, rather than aggregating these hypotheses, the continuous flow network is explicitly trained to render the specific observed reality. To achieve this, we isolate the Gaussian prior corresponding to the winning hypothesis k^*. A base trajectory for the future horizon, $\mathbf{x}_{0,f}$, is sampled directly from this mode-specific distribution:

$$\mathbf{x}_{0,f} \sim \mathcal{N}(\mu_{k*}, \sigma_{k*}^2) \tag{7}$$

To stabilize the training, we apply a stop-gradient operator; The flow matching network is conditioned on the discrete mode, but gradients are blocked from flowing back into both the Hypothesis Generator and the Context Encoder ($\text{sg}(\mu_{k^*})$ and $\text{sg}(\mathbf{h}_{ctx})$).This explicitly isolates the first stage for structural regime discovery from the second stage for local density estimation, preventing moving targets during joint optimization.

2.5. Continuous Residual Refinement via Flow Matching

During training, a random time step $t \sim \mathcal{U}(0,1)$ is sampled. The intermediate state $\mathbf{x}_t$ is constructed by concatenating the known past $\mathbf{y}_p$ with a linear interpolation of the future horizon. This effectively masks the past from the flow process, treating it as a fixed prefix: $\mathbf{x}_t = \mathbf{y}_p \oplus (t\mathbf{y}_f + (1-t)\mathbf{x}_{0,f})$. The velocity field $v_\theta(\mathbf{x}_t, t, \mathbf{c})$ is parameterized using a deep network of Gated Residual Blocks similar to the Diffwave architecture (Kong et al., 2021). The conditioning vector $\mathbf{c}$ contains the covariates $\mathbf{C}$, the detached global context $\text{sg}(\mathbf{h}_{ctx})$, a learned embedding of the winning mode index k^*, and a reference trajectory formed by concatenating the known past demand with the detached winning hypothesis, $\mathbf{y}_p \oplus \text{sg}(\mu_{k^*})$. The network is trained to regress the target velocity of the future horizon, $u_{target} = \mathbf{y}_f - \mathbf{x}_{0,f}$. To ensure the model only updates the prediction horizon while remaining conditioned on the fixed past, we define $v_\theta^{(future)}$ as the slice of the predicted vector field corresponding to the future F time steps. The CFM loss is thus defined as:

$$\begin{aligned}\mathcal{L}_{CFM} = \mathbb{E}_{t,\mathbf{x}_{0,f},\mathbf{y}_f}\Big[\Big\| v_\theta^{(future)}(\mathbf{x}_t, t, \mathbf{c}) \\ - (\mathbf{y}_f - \mathbf{x}_{0,f})\Big\|_2^2\Big]\end{aligned} \tag{8}$$

2.6. Inference

Given past demand $\mathbf{y}_p$, covariates $\mathbf{C}$, and building metadata $\mathbf{m}$ the process begins by extracting the global context $\mathbf{h}_{ctx}$ via the Context Encoder. The Probability Head evaluates $\mathbf{h}_{ctx}$ to output the occurrence probabilities π for all K possible future regimes. Simultaneously, the Hypothesis Generator provides the structural parameters $\{(\mu_k, \sigma_k^2)\}_{k=1}^K$. These parameters dynamically define the hypothesis-specific base distribution for the flow model. For each scenario, a base future trajectory $\mathbf{x}_{0,f}$ is drawn from $\mathcal{N}(\mu_k, \sigma_k^2)$. An Euler solver is then used to sequentially integrate the predicted velocity field v_θ from $t = 0$ to $t = 1$, resulting in the probabilistically weighted forecast $\mathbf{x}_1$ for that specific hypothesis.

3. Experiments

3.1. The Aalborg Smart Meter Dataset

The Aalborg Smart Heat Meter Dataset is a public repository containing district heating consumption records from residential buildings in Aalborg, Denmark (Schaffer et al., 2022). The dataset spans a three-year period from January 2018 to December 2020 at an hourly temporal resolution. It includes a preprocessed subset of 3,021 meters, following the exclusion of buildings with ambiguous metadata. We pair the consumption time-series with two distinct feature sets: Static Building Metadata including categorical and registration data regarding building properties, specifically dwelling type (categorized as apartments, terraced houses, single-family houses and non residential buildings), construction year, and energy level ratings, and External temperature records, collected from a nearby station in Aalborg, sourced via the Danish Meteorological Institute (DMI) [1].

3.2. Training

To evaluate the proposed framework against recent work that used the same dataset (Chen et al., 2023; Qian et al., 2024), we choose the same past window of $P = 45$ days to forecast across three daily horizons: 3, 5, and 7 days. The dataset is chronologically partitioned into training (70%), validation (10%), and test (20%) splits. Temporal covariates include ambient temperature and calendar features (hour of day, day of week) for schedule-based consumption patterns. Furthermore, we incorporate a learnable building-specific embedding to separate individual trajectories and include the building type (e.g., single-family houses, non-residential buildings) and the construction year. The MC-Flow framework is trained end-to-end using the AdamW optimizer with a learning rate of 10^{-3}. We choose $K = 16$ discrete structural modes (hypotheses). We found that values around $K = 16$ provide the optimal trade-off between mode coverage and computational overhead during inference. Specifically, decreasing the number of hypotheses limits the model's expressivity. Conversely, increasing K yields diminishing returns as redundant hypotheses tend to converge toward identical targets, increasing the inference overhead without a significant improvement of the resolution of the predictive distribution

3.3. Evaluation

3.3.1. Baselines

To evaluate the performance of our proposed framework, we benchmark it against deterministic, probabilistic, and generative models. MSDGN (Huang et al., 2025), designed for multi-step heat load forecasting, captures temporal patterns via a multi-scale dynamic graph structure. We adopt it as our primary deterministic baseline as it demonstrated the best performance on this exact dataset against an extensive suite of architectures, including GAIN (Zhang et al., 2024), MAGNN (Chen et al., 2023), and Transformers. For standard probabilistic forecasting, we compare against DeepAR (Salinas et al., 2019), an autoregressive model that parameterizes a global likelihood distribution using recurrent neural networks. To evaluate our structured prior against flow matching approaches, we compare against TS-Flow (Kollovieh et al., 2025). TSFlow utilizes Conditional Flow Matching driven by a Gaussian Process prior conditioned on past context. We implement TSFlow using its

[1] meteorological data available via: https://www.dmi.dk/

proposed generic periodic kernel, generating 160 samples and using 16 Euler steps. Finally, we evaluate an MCL-Only discrete hypothesis ablation. By removing the continuous flow matching refinement step, this variant relies only on the learned hypotheses.

3.3.2. Evaluation Metrics

To evaluate structural accuracy and distributional calibration, we employ the **Weighted CRPS** as formulated in (Dai et al., 2026). This metric provides a unified framework for assessing both discrete hypotheses and uniform stochastic samples. Given a ground truth trajectory $\mathbf{y} \in \mathbb{R}^F$, a set of N generated scenarios $\{\mathbf{x}_n\}_{n=1}^N$, and a corresponding probability vector $p = (p_1, \dots, p_N)$, the score is:

$$\text{CRPS} = \sum_{n=1}^{N} p_n \|\mathbf{x}_n - \mathbf{y}\|_1 - \frac{1}{2} \sum_{n=1}^{N} \sum_{j=1}^{N} p_n p_j \|\mathbf{x}_n - \mathbf{x}_j\|_1. \tag{9}$$

To ensure a consistent comparison, we fix the total scenario budget for probabilistic models at $N = 160$. For the MCL-Only ablation, the metric is computed over the $K = 16$ discrete anchors μ_k weighted by their predicted probabilities π_k. For DeepAR and TSFlow, we draw $N = 160$ samples and assign uniform weights $p_n = 1/N$. For MC-Flow, the $N = 160$ scenarios are distributed across the K structural regimes, yielding $S = 10$ refined samples per mode, each assigned a weight of $p_n = \pi_k/S$, marginalizing the local refinement over the discrete mode prior. We also evaluate point-forecast accuracy via **RMSE and MAE**. For the MCL-Only ablation, this is the deterministic mode μ_k associated with the maximum predicted probability $\arg\max_k \pi_k$. For MC-Flow, the point estimate is the mean of the S samples belonging to this highest-probability regime. Conversely, for baselines lacking a structured discrete prior (DeepAR and TSFlow), the global mean of all N generated samples is adopted as the representative point forecast.

3.3.3. Predictive and Probabilistic performance

We evaluate the model using three different forecast horizons (3 days, 5 days and 7 days). Best results are highlighted in **bold**, and second-best are underlined.

Table 1. Forecasting performance

Model	**RMSE (↓)**			**MAE (↓)**			**CRPS (↓)**		
	3	5	7	3	5	7	3	5	7
MSDGN	8.143	8.675	8.679	5.458	5.844	5.707	–	–	–
DeepAR	9.144	10.110	10.753	6.122	6.961	6.994	0.131	0.148	0.151
TSFlow(16-Euler)	**6.329**	**6.939**	**7.106**	**4.041**	**4.442**	4.627	0.092	**0.102**	**0.105**
TSFlow(1-Euler)	6.674	7.186	7.144	4.217	4.461	4.632	0.110	0.122	0.120
MCL-Only	6.652	7.214	7.492	4.189	4.594	4.817	0.087	0.117	0.124
MC-Flow(1-Euler)	6.645	7.129	7.238	4.162	4.545	**4.576**	**0.085**	0.106	**0.105**

As shown in Table 1, MC-Flow outperforms the deterministic baseline MSDGN and DeepAR across all horizons. This performance gap suggests that strict Gaussian assumptions are insufficient for capturing the multimodal nature of building-level heat demand. While TSFlow (16-Euler) achieves marginally lower RMSE by using more integration steps, MC-Flow maintains competitive CRPS scores using only a single Euler step. We attribute this to the hierarchical design: by using discrete modes as an explicitly structured prior, the transport map is initialized closer to the target distribution. This results in inherently straighter probability paths that require minimal integration steps to characterize local uncertainty. The comparison with the *MCL-Only* ablation shows the role of the flow-matching stage. While the discrete structural modes provide a robust initial forecast at $h = 3$ (CRPS=0.087), performance degrades at $h = 7$ (CRPS=0.124). Because the hypotheses are optimized via an MSE-based objective, they yield inherently smooth trajectories that act only as coarse approximations. As temporal variance increases over longer horizons, the flow-matching refinement becomes beneficial for accurate density estimation.

The primary contribution of our approach lies in its operational interpretability and flexibility. Standard generative models produce an ensemble of stochastic samples that are difficult for grid operators to translate into actionable decisions. In contrast, our hierarchical approach explicitly decouples structural uncertainty from local sampling. This enables on-demand sampling. Although our quantitative evaluation used $S = 10$ samples per mode for baseline parity, the MC-Flow design allows operators to isolate a specific regime of interest in practice. An operator can select a single hypothesis and sample from the flow model to quantify the localized variance exclusively for that scenario, providing targeted insights without the computational overhead of sampling the entire joint distribution. Further qualitative results and visualizations of these regime-specific samples are provided in Appendix G.

4. Conclusion

This work addresses the inherent stochasticity of heat demand by proposing a novel hierarchical framework that bridges two distinct probabilistic paradigms: discrete hypothesis learning via Multiple Choice Learning and continuous refinement via Flow Matching. For district heating operators, this framework provides practical utility by generating actionable scenarios and enabling on-demand sampling within specific operational regimes. A limitation of this approach is that poor initialization during the MCL stage can constrain the subsequent Flow Matching refinement. Future research will further explore the interplay between these two paradigms. We will also focus on integrating the approach with downstream stochastic optimization frameworks to drive automatic resilient grid operations.

Impact Statement

This paper presents work whose goal is to advance the field of Machine Learning. There are many potential societal consequences of our work, none which we feel must be specifically highlighted here.

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A. Visual abstract

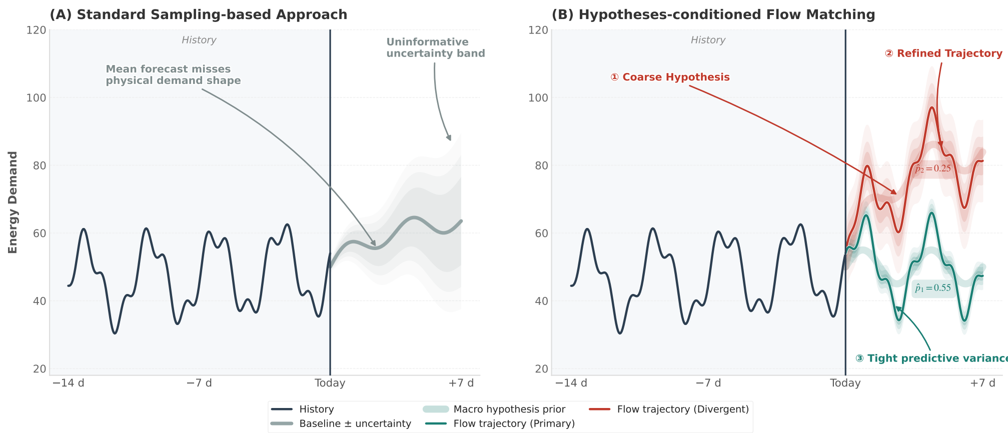


Figure 1. Visual comparison of a standard probabilistic forecast versus the proposed multi-hypothesis framework. Standard approaches often yield a single, uninformative uncertainty cone. In contrast, our approach explicitly separates aleatoric uncertainty into distinct, probabilistically weighted demand trajectories, providing operators with actionable scenarios.

B. Example of two distinct demand profiles

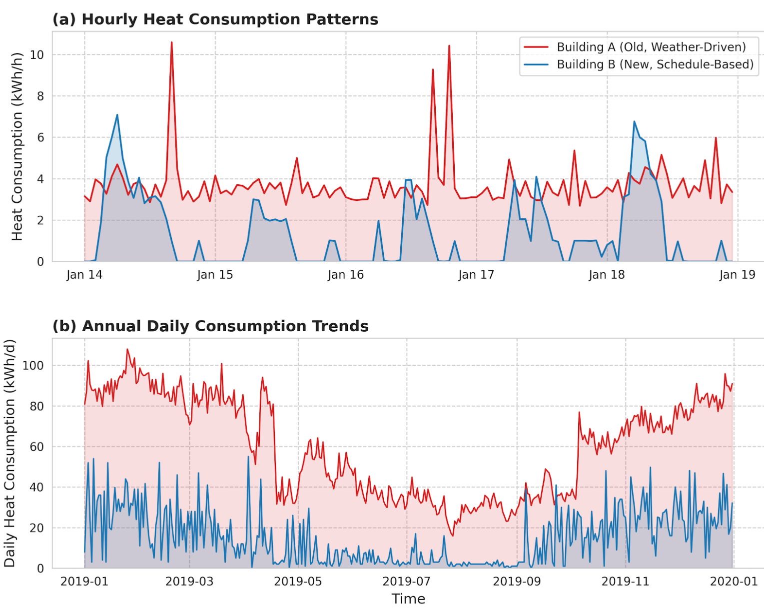


Figure 2. Visual comparison of two distinct demand profiles. Some buildings are weather-driven while others, more modern have manually adjusted load.

C. Preprocessing

The raw target variable represents the daily heat energy consumption measured in kWh. We apply a two-step transformation:

1. **Log-Plus-One Transform:** We first apply $y' = \log(1 + y_{raw})$ to account for peaks while maintaining the zero lower-bound of demand.
2. **Building-Specific Standardization:** The final normalized target is computed as $y = (y' - \mu^{(b)}_{train})/\sigma^{(b)}_{train}$.

During inference, generated scenarios are inverted through this exact pipeline via an exponential transform ($\exp(y \cdot \sigma^{(b)}_{train} + \mu^{(b)}_{train}) - 1$) before calculating the physical evaluation metrics.

C.1. Exogenous Covariates and Metadata

The model is conditioned on both dynamic temporal covariates and static building metadata:

- **Dynamic Covariates (C):** The temporal features include standard normalized daily temperature and cyclic calendar features (sine and cosine encodings for both the day-of-week and day-of-year). Additionally, a binary observation mask distinguishes between the past look-back window (1) and the future horizon (0).

- **Static Metadata:** Building-specific profiles are constructed using a categorical Building ID, a categorical Unit Type, and a continuous Construction Year feature. The construction year is normalized to a $[0, 1]$ scale via $(year - 1900)/100$, with missing values imputed using dataset median.

D. Vizualization of the MC-Flow Framework

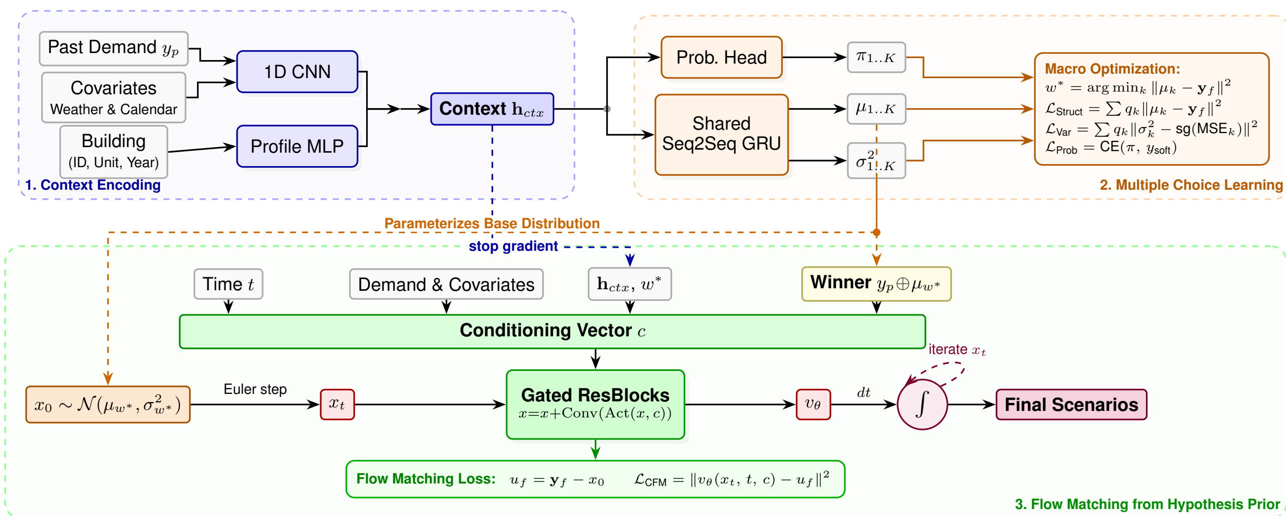


Figure 3. Overview of the MC-Flow architecture. The Context Encoder extracts a global summary to first generate K discrete demand hypotheses via Multiple Choice Learning. The winning hypothesis then dynamically parameterizes a base distribution for Conditional Flow Matching.

E. Network Architecture

The framework is implemented in PyTorch. All linear and convolutional layers employ SiLU activation functions unless otherwise specified.

The Context Encoder merges the dynamic time series and the static metadata into a global context vector $\mathbf{h}_{ctx} \in \mathbb{R}^{160}$.

- **Temporal CNN:** The $P \times 7$ input sequence (past demand concatenated with 6 dynamic covariates) is processed by a 3-layer 1D Convolutional Neural Network. The sequence of channels is $32 \to 64 \to 128$. The first layer uses a kernel size of 5 (padding 2), while subsequent layers use a kernel size of 3 (padding 1). Each convolution is followed by 1D Batch Normalization, a SiLU activation, and a Max Pooling layer with stride 2. The temporal dimension is subsequently pooled using an Adaptive 1D Average Pooling layer, yielding a 128-dimensional temporal embedding.

- **Profile MLP:** The categorical Building ID and Unit Type are passed through learned embedding layers of dimension 16 and 8, respectively. These are concatenated with the continuous construction year and processed by a 2-layer MLP (hidden dimension 32) to output a 32-dimensional static profile vector.

E.1. Multiple Choice Learning

- **Probability Head:** $\mathbf{h}_{ctx}$ is processed by a 2-layer MLP with a hidden size of 128 to output $K = 16$ raw logits.
- **Hypothesis Generator:** The structural regime shapes μ_k and variances σ_k^2 are generated using a SeqtoSeq Gated Recurrent Unit (GRU). The GRU uses a single layer with 128 hidden units. Its initial hidden state is generated by passing $\mathbf{h}_{ctx}$ through a linear projection. The outputs are linearly projected to the target dimension D, and a Softplus activation gives the variance.

To prevent the Probability Head from converging to a mode-agnostic mean before the Hypothesis Generator has sufficiently separated the structural regimes, we apply a linear warmup to the weight of the Cross-Entropy loss ($\mathcal{L}_{\text{Prob}}$) over the first 10 epochs.

E.2. Continuous Flow Matching Stage

The optimal transport vector field v_θ is parameterized using a deep network of 5 Gated Residual Blocks, drawing inspiration from the DiffWave architecture.

- **Conditioning Vectors:** The time integration variable t is encoded using a sinusoidal positional embedding projected to 32 dimensions. The global context $\mathbf{h}_{ctx}$ and the detached structural reference trajectory $\mathbf{y}_p \oplus \text{sg}(\mu_{k^*})$ are projected to 64 and 32 dimensions respectively. Along with the projected dynamic covariates, the total conditioning dimension passed to the residual blocks is 160.
- **Gated Residual Blocks:** Each of the 5 blocks maintains 64 residual channels. To ensure a large temporal receptive field the blocks use exponentially increasing 1D convolution dilations $d \in \{1, 2, 4, 8, 16\}$.

E.3. Training algorithm

Algorithm 1 Training

Input: Training dataset $\mathcal{D} = \{(\mathbf{y}_p, \mathbf{C}, \mathbf{m}, \mathbf{y}_f)\}$, number of modes K, network parameters θ, total epochs E.

for epoch $e = 1, \dots, E$ **do**
  for each batch $(\mathbf{y}_p, \mathbf{C}, \mathbf{m}, \mathbf{y}_f) \in \mathcal{D}$ **do**
    ▷ **Stage 1: Discrete Regime Discovery**
    $\mathbf{h}_{ctx} \leftarrow \text{ContextEncoder}(\mathbf{y}_p, \mathbf{C}, \mathbf{m})$
    $\{\mu_k, \sigma_k^2\}_{k=1}^K \leftarrow \text{HypothesisGenerator}(\mathbf{h}_{ctx}, \mathbf{C}_f)$
    $\pi \leftarrow \text{ProbabilityHead}(\mathbf{h}_{ctx})$
    $k^* \leftarrow \arg\min_k \|\mu_k - \mathbf{y}_f\|_2^2$
    $y_{soft} \leftarrow \text{Softmax}(-\|\mu_k - \mathbf{y}_f\|_2^2 / \tau_{target})$
    Compute $\mathcal{L}_{\text{discrete}}$ using k^*, π, and y_{soft}
    ▷ **Stage 2: Conditional Flow Matching**
    $\mathbf{x}_{0,f} \sim \mathcal{N}(\text{sg}(\mu_{k^*}), \text{sg}(\sigma_{k^*}^2))$
    $t \sim \mathcal{U}(0, 1)$
    $\mathbf{x}_t \leftarrow \mathbf{y}_p \oplus (t\mathbf{y}_f + (1-t)\mathbf{x}_{0,f})$
    $\mathbf{c} \leftarrow \text{Concat}(\mathbf{C}, \text{sg}(\mathbf{h}_{ctx}), \text{sg}(\mu_{k^*}), \text{embed}(k^*))$
    $\mathcal{L}_{CFM} \leftarrow \|v_\theta^{(future)}(\mathbf{x}_t, t, \mathbf{c}) - (\mathbf{y}_f - \mathbf{x}_{0,f})\|_2^2$
    ▷ **Joint Optimization**
    Update parameters θ using $\nabla_\theta(\mathcal{L}_{\text{discrete}} + \mathcal{L}_{CFM})$
  end for
end for

Output: Optimized model parameters θ^*

F. Probability Calibration

The calibration of the predictive distributions is evaluated via the Reliability Diagram and the Probability Integral Transform (PIT) Histogram Figure 4. The Reliability Diagram demonstrates that the model's predicted probabilities align closely with the empirical frequencies, indicating near-perfect empirical coverage across the test set. Furthermore, the PIT histogram exhibits a high degree of uniformity indicating that the model neither systematically underestimating nor overestimating the uncertainty of heat demands.

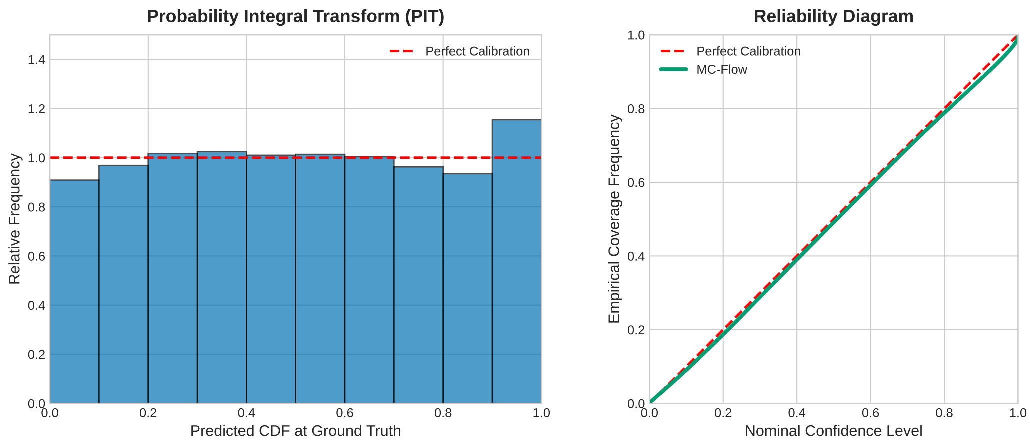
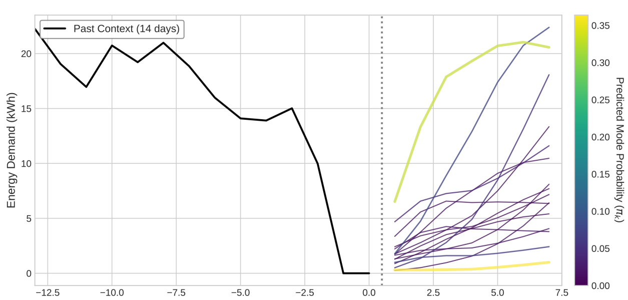


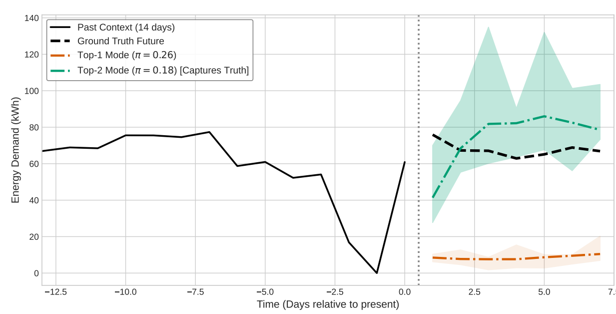
Figure 4. PIT Histogram and Reliability Curve

G. Qualitative Analysis

As shown in Figure 5, the model outputs distinct macro-level hypotheses alongside explicit occurrence probabilities (Figure 5a). Figure 5b shows two divergent modes and their associated local uncertainty. Notably, even when the ground truth aligns with a lower-probability trajectory, the model captures the event as a distinct hypothesis. Figure 6, presents random forecast windows from the test set for a given building. The model can characterize rare events as plausible, lower-probability scenarios.



(a) Explicitly assigned probabilities



(b) Local uncertainty via flow matching.

Figure 5. Hierarchical predictive distribution: (a) categorical probability separating regimes, and (b) local variance rendering around modes.

Figure 7 depicts the top four trajectories for consecutive forecasting windows. Significant divergence is observed around day 120, where the model identifies a potential decrease in demand as a separate structural mode. Such rolling view provides operators with trackable scenarios facilitating more reliable planning and network balancing.

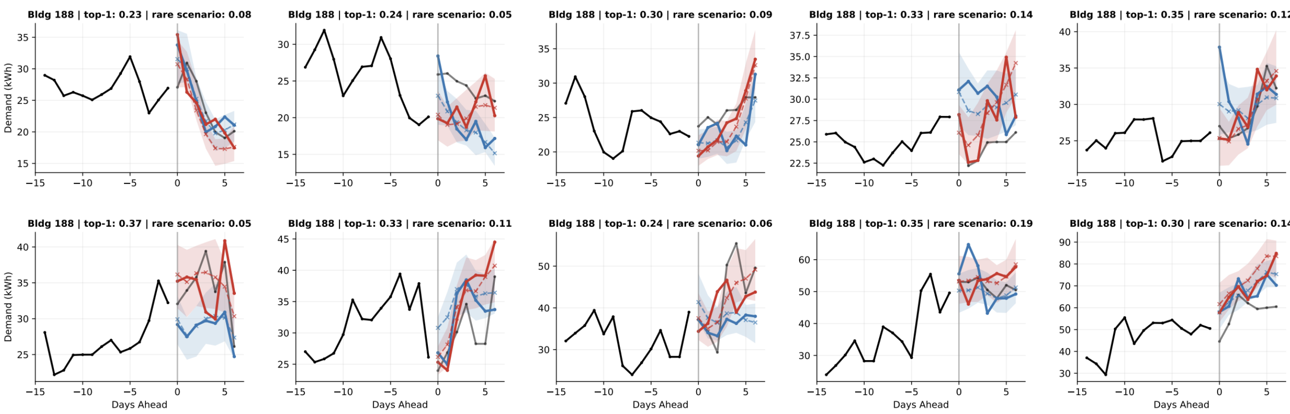


Figure 6. Random forecast test windows for a given building showing top-1 scenario (blue) and a low probability scenario (red)

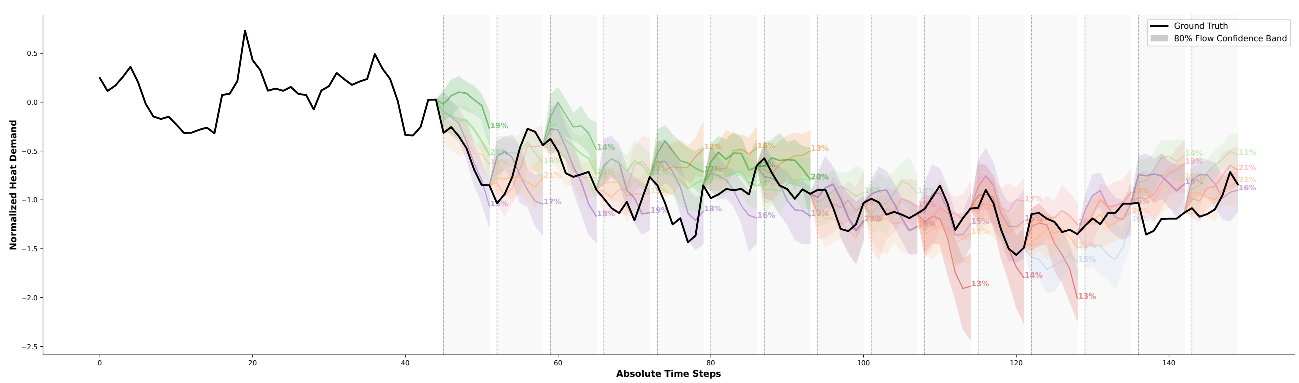


Figure 7. Example of a rolling forecast showing the top four trajectories and their associated hypotheses probabilities.